Lina Stensland, Anne Jumper, Sheelyn Pharr
Dr. Erica Kemmerling
ME51 Thermal Fluid Systems II
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Zambia Irrigation System Simulation
Global Perspectives Final Project

Introduction

Agriculture is a huge part of the Zambian economy, and the majority of people in Zambia work in agriculture. The country has 1.6 million farms which are 5 acres or less, meaning they are often small-scale operations run by families or single people. Zambia has been experiencing a drought since January of 2024, which is an issue for a country reliant on rain-fed agriculture. This has had huge impacts on food production and availability, with the drought being declared a national disaster in February of 2024. Families don't have enough crops to sell for profit or to even eat, leaving them reliant on aid organizations to survive.[1] Women especially are affected, with many facing violence and malnutrition due to the drought's wide reaching effects. Some women trade sex to procure food and water for their families. While there are education systems in place to support teen girls and educate them on the risks of transactional sex and teen pregnancy, people will also target women and female-led households for exploitation, violence, and abuse. With this drought, Zambian citizens also face water shortages, leaving many communities with no access to clean drinking water. This means any agricultural irrigation system must be as water-efficient as possible. Beyond families, aid organizations and the Zambian government are involved in the issue. They are in charge of policy, water regulations, and aid distributions.

In this problem, the team is hired by Chisha Folotiya, a local Zambian businessman and farmer who is interested in agricultural infrastructure and management.[2] We are tasked with designing a drip irrigation system for eggplants being grown on his property. Drip irrigation systems are low maintenance for the farmers and provide a steady drip of water to plants directly at their roots. Due to the shortage of water, drip tape systems are a great choice, since they provide focused water point by point to the plants. These systems can be automated with soil moisture sensors or timers if a farmer desires. This predominantly hands-off method of providing continuous water to plants is good for these small-scale farms, since farmers only have so much work capability. These drip tape systems only require maintenance and cleaning debris from the water filter attached. Folotiya also details more benefits of drip irrigation for farmers in a video posted to his YouTube channel.[3] Finally, drip irrigation is cheap. Many of the Zambian farmers hit hard by the drought do not have the money for expensive irrigation systems. The financial

[1] https://www.unocha.org/news/seeds-hope-zambias-fight-against-drought
[2] https://mondofarms.co.zm/about-us-2/
[3] https://www.youtube.com/watch?v=Blb0_8ZcMwc

accessibility of a drip tape irrigation system is key for farmers' ability to adopt the system for their own farms.

Analysis

1) Field and column flow requirements

150 plants/column * 1.2×10^{-3} m^3 water per plant per day = each column needs 0.18 m^3 of water per day

That is 0.0075 m^3 per hour or 2.0833×10^{-6} m^3/s for a single column (drip tape)

For a single column:

$$150\frac{plants}{column} \times 1.2 \times 10^{-3}\frac{m^3\ water}{plant\ per\ day} = 0.18\frac{m^3}{day} \times \frac{1\ day}{24\ hr} \times \frac{1\ hr}{3600\ s} = 2.0833 \times 10^{-6}\ m^3/s$$

150 columns in the field, entire field needs 3.125×10^{-4} m^3/s (PVC)

The flow requirements for a column are the same as a row because they both contain 150 plants. The design will primarily use columns, so this flow rate is defined as flow for a column to maintain consistency for the rest of the report.

2) Design

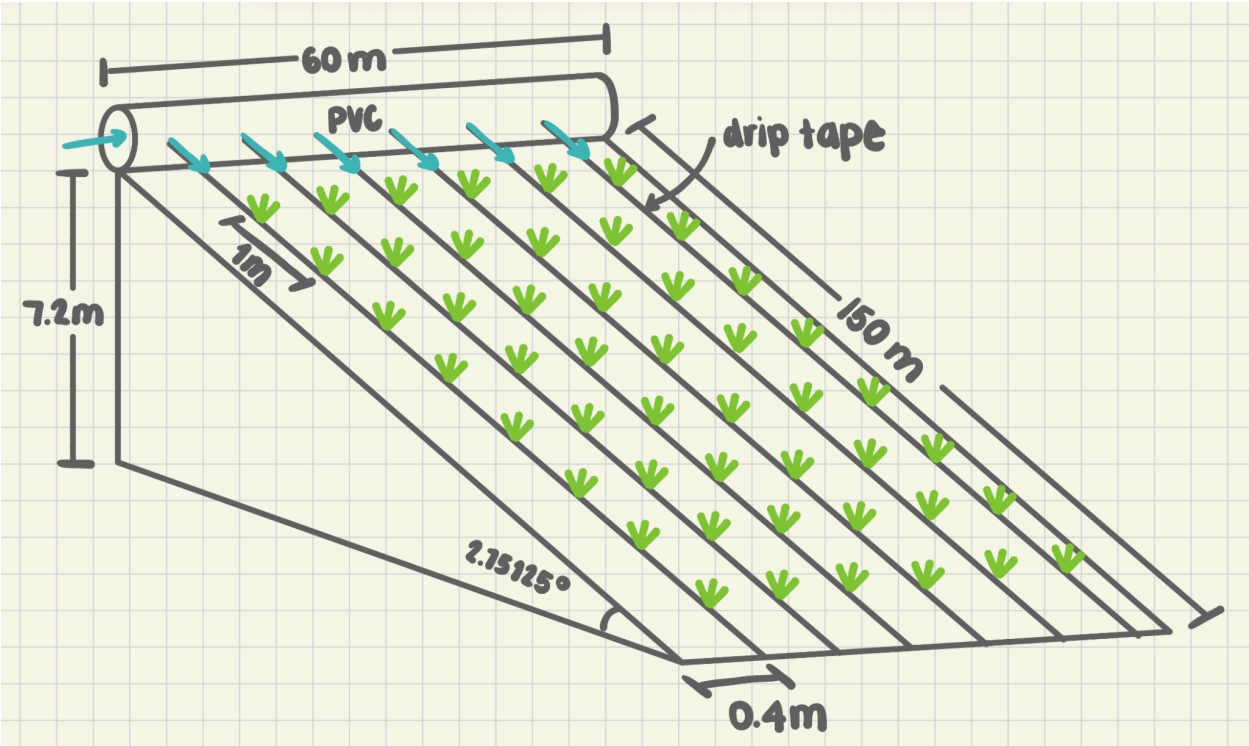


Figure 1: *Drip Tape Irrigation design.*

There is one PVC pipe along the top of the field with diameter 0.05 m extending from the inlet. There are 150 pieces of drip tape coming off of the PVC pipe at 90 degree angles with 0.4 meter spacing (3 pictured in *Figure 1*). The drip tape has a diameter of 0.0127 m and side-facing holes with diameters of 0.002m. Each piece of drip tape services one column and 150 plants (8 pictured in *Figure 1*).

3) Head loss of system to First and Last Plant

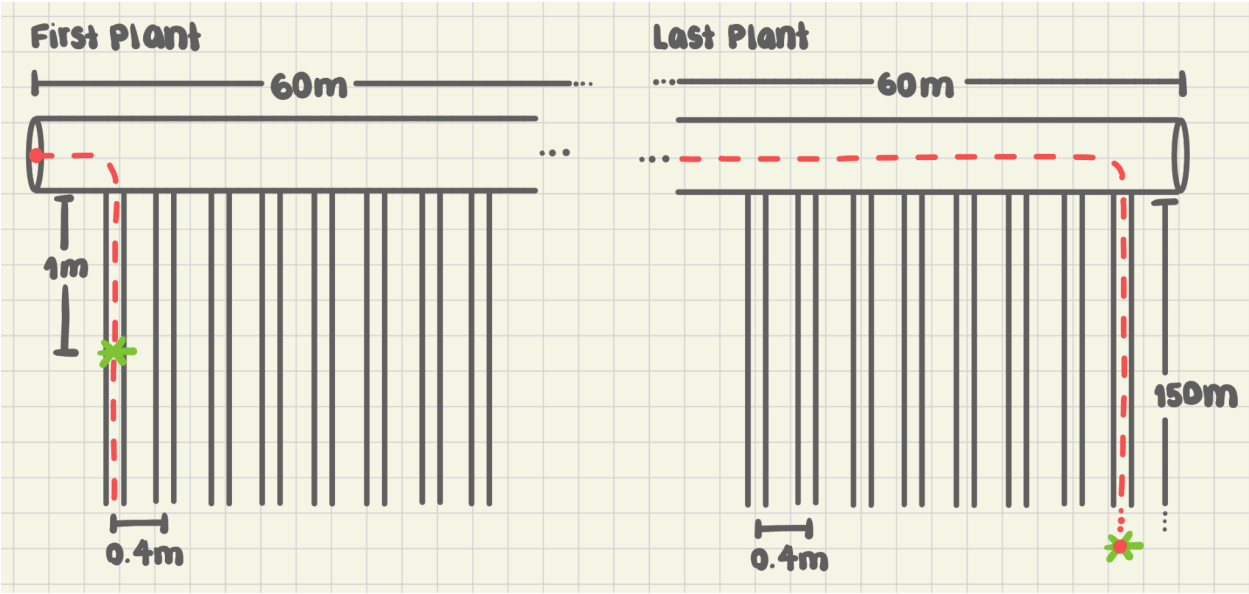


Figure 2: *Streamline to the first plant and last*

First Plant Head Loss:

Head losses for the first plant include, minor loss from the turn transition from PVC to drip tape and major losses in the PVC pipe and the drip tape leading up to the plant. Inlet flow rates and corresponding velocities were used because change in velocity is negligible over the small distance.

Inlet speeds based on necessary flow rate:

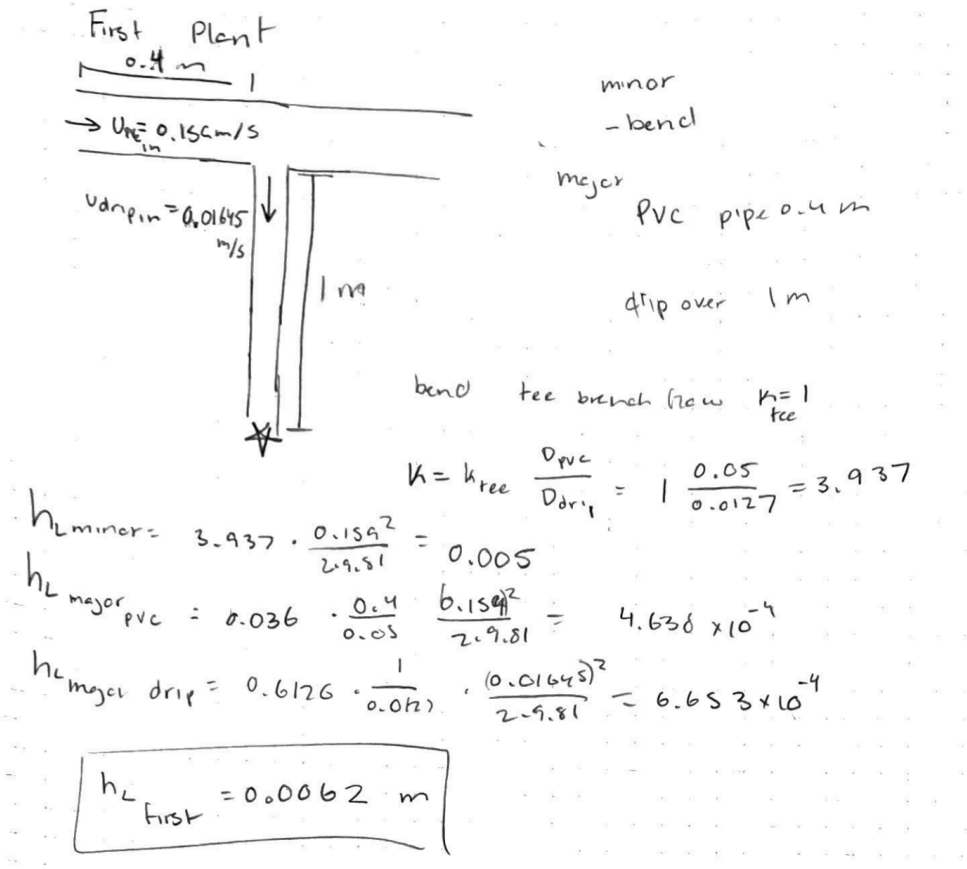
$$V = \frac{Q}{A}$$

PVC inlet speed

$$U_{pvc\ inlet} = \frac{3.125 \times 10^{-4}\ m^3/s}{\frac{\pi}{4}(0.05)^2} = 0.159\ m/s$$

$$U_{drip\ inlet} = \frac{2.0833 \times 10^{-6}}{\frac{\pi}{4}(0.0127)^2} = 0.01645\ m/s$$

Minor and major head losses were calculated individually and summed for total head loss:



Last Plant Head Loss:

There are major and minor components of the head loss for the streamline to the last plant. Major head loss components for the last plant include major head loss from the PVC and the drip tape. One form of minor loss was inline minor loss for both the PVC and drip tape resulting from the multiple branches and diversions of flow (drip tape diverting flow from PVC pipe, and watering holes diverting flow from the drip tape). The other minor head loss was at the turn from PVC to drip tape. For the major loss and inline minor loss calculations average velocities along the pipe were used. These were calculated by averaging the starting and ending velocities of the PVC and drip tape sections. A modified version of Bernoulli's equation was used to calculate the end velocity of the PVC flow based on the starting velocity for the drip tape derived from the required flow rate.

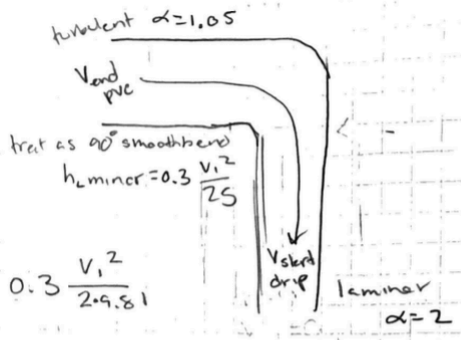
Last plant Headloss

minor: inline loss from drip tape, turn (elbow)

major: PVC length, drip tape length

find U_{avg} PVC

$$\frac{V_{start\ PVC} + V_{end\ PVC}}{2}$$



$V_{start\ pipe} = 0.01645$ m/s

$$1.05 \frac{V_1^2}{2 \cdot 9.81} = 2 \frac{0.01645^2}{2 \cdot 9.81} + 0.3 \frac{V_1^2}{2 \cdot 9.81}$$

$$0.75 \frac{V_1^2}{2 \cdot 9.81} = 2 \frac{(0.01645)^2}{2 \cdot 9.81}$$

$$V_1^2 = \frac{2}{0.75} (0.01645)^2$$

$$\sqrt{V_1^2} = \sqrt{7.2161 \times 10^{-4}}$$

$$V_{end\ PVC} = V_1 = 0.02686$$

$$U_{avg\ PVC} = \frac{0.159 + 0.02686}{2} = \boxed{0.09293 \text{ m/s}}$$

$$U_{avg\ drip} = \frac{0.01645 + 0}{2} = \boxed{0.008225 \text{ m/s}}$$

Reynolds #s = $\frac{DU_{avg}}{\nu}$ $\nu = 1\times10^{-6}\ m^2/s$

$Re_{pvc} = \frac{0.05 \cdot 0.09293}{1\times10^{-6}} = 4646.5 > 2000$ turbulent

$Re_{drip} = \frac{0.0127 \cdot 0.008225}{1\times10^{-6}} = 104.45$ $Re_{drip} < 2000$ laminer

$\varepsilon = 0.0015\ mm$

$e/D = \frac{0.0015\times10^{-3}}{0.0127} = 1.18\times10^{-4}$

moody $f_{pvc} \approx 0.036$

$f_{drip} = \frac{64}{Re} = 0.6126 = f_{drip}$

$h_{L_{pvc\ mayor}} = 0.036 \cdot \frac{160}{0.05} \cdot \frac{(0.0859)^2}{2g} = 0.0163m$

$h_{L_{drip\ mayor}} = 0.6126 \cdot \frac{150}{0.0127} \cdot \frac{(0.008225)^2}{2g}$

$h_{L_{drip\ mayor}} = 0.02495\ m$

$h_{L_{mayor}} = 0.0163 + 0.02495 = 0.041215\ m$ [4]

minor loss

inline loss pvc

estimate for a smaller branch ϕ

$K_L = K_{L\ tee\ inline} \cdot \frac{D_{drip}}{D_{pvc}} = 0.2\ \frac{0.0127}{0.05} = 0.0508$

inline loss $= K_L \cdot \frac{U_{avg}^2}{2g} \cdot 149$

$h_{L_{inline\ pvc}} = 149 \cdot 0.0508\ \frac{0.09293^2}{2g} = 0.00333\ m$

bend for last pipe

$h_{L_{minor\ bend}} = 1.1\ \frac{0.012928^2}{2g} = 9.37\times10^{-6}\ m$

$h_{L_{minor}} = 9.37\times10^{-6} + 0.00333 = 0.003339\ m$

inline loss drip

$K_L = K_{L\ tee\ inlin} \cdot \frac{D_{hole}}{D_{drip}} = 0.2 \cdot \frac{0.002}{0.0127} = 0.03149$

$h_{L_{inlinedrip}} = 149 \cdot 0.03149 \cdot \frac{0.008225^2}{2\cdot9.81} = 1.618\times10^{-5}$

$h_{L_{minor}} = h_{L_{inlinedrip}} + h_{L_{inline\ pvc}} + h_{L_{bend}} = 0.00333 + 9.37\times10^{-6} + 1.618\times10^{-5}$

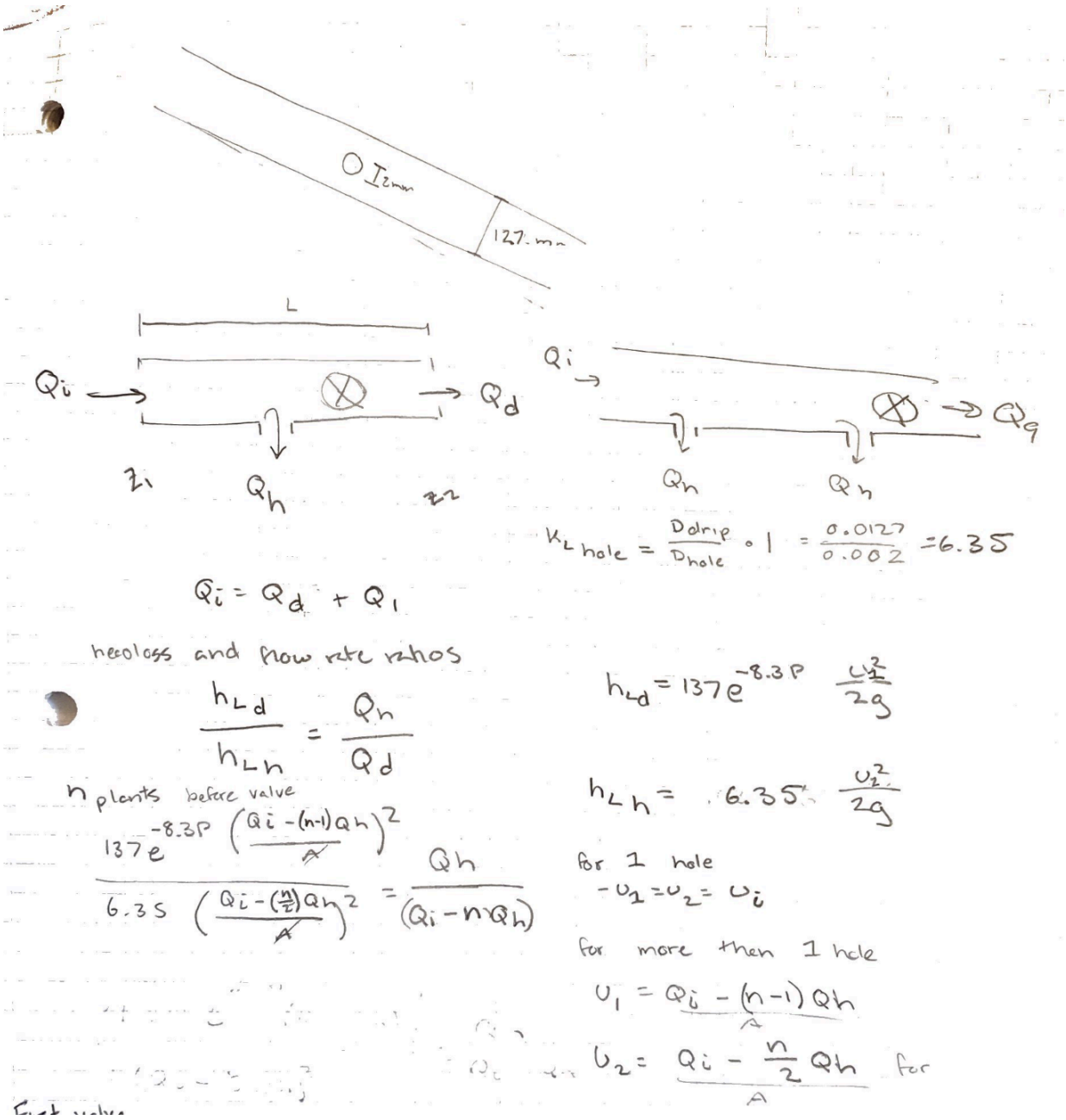
$h_{L_{minor}} = 0.003356$

$h_{L_{last}} = h_{L_{minor}} + h_{L_{mayor}} = 0.003356 + 0.04121 = 0.04457\ m$

[4] https://www.ti-soft.com/en/support/help/heatingdesign/project/parameters/hydronic-settings/pipe-roughness/absolute_pipe_roughness

4) Valves

Valves were put in drip tape 15 meters apart. It is unrealistic to put a valve after every plant. We spaced 9 valves out across the column, so that there would be additional flow control throughout. The first valve is after the 15th plant in the column, 15 m after the PVC to drip tape bend and the last one is 15 m before the end of the drip tape.



$$Qinlet := 2.0833 \cdot 10^6$$

$$Qh := \frac{Qinlet}{150} = 1.389 \cdot 10^4$$

valves 30 plants

valve every 15

first valve

$$n := 15$$

$$Qi := Qinlet = 2.083 \cdot 10^6$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.94981044137549972608$$

second valve

$$Qi := Qinlet - n\ Qh = 1.875 \cdot 10^6$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.93426099008458930024$$

third valve

$$Qi := Qinlet - 2\ n\ Qh = 1.667 \cdot 10^6$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.91644074795376297 71$$

fourth valve

$$Qi := Qinlet - 3\ n\ Qh = 1.458 \cdot 10^6$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.89558434220239600758$$

fifth valve

$$Qi := Qinlet - 4\ n\ Qh = 1.25 \cdot 10^6$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.87046799040376225292$$

sixth valve

$$Qi := Qinlet - 5\ n\ Qh = 1.042 \cdot 10^6$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.83895942261870838683$$

seventh valve

$$Qi := Qinlet - 6\ n\ Qh = 8.333 \cdot 10^5$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.79684880198516606623$$

eight valve

$$Qi := Qinlet - 7\ n\ Qh = 6.25 \cdot 10^5$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.73397783585581232973$$

ninth valve

$$Qi := Qinlet - 8\ n\ Qh = 4.167 \cdot 10^5$$

$$\frac{\left(137\ e^{-8.3\,P}\ \left((Qi-(n-1)\ Qh)^2\right)\right)}{\left(6.35\left(Qi-\left(\frac{n}{2}\right)\ Qh\right)^2\right)} = \frac{Qh}{(Qi-n\ Qh)} \xrightarrow{solve} 0.61418351085356005 17$$

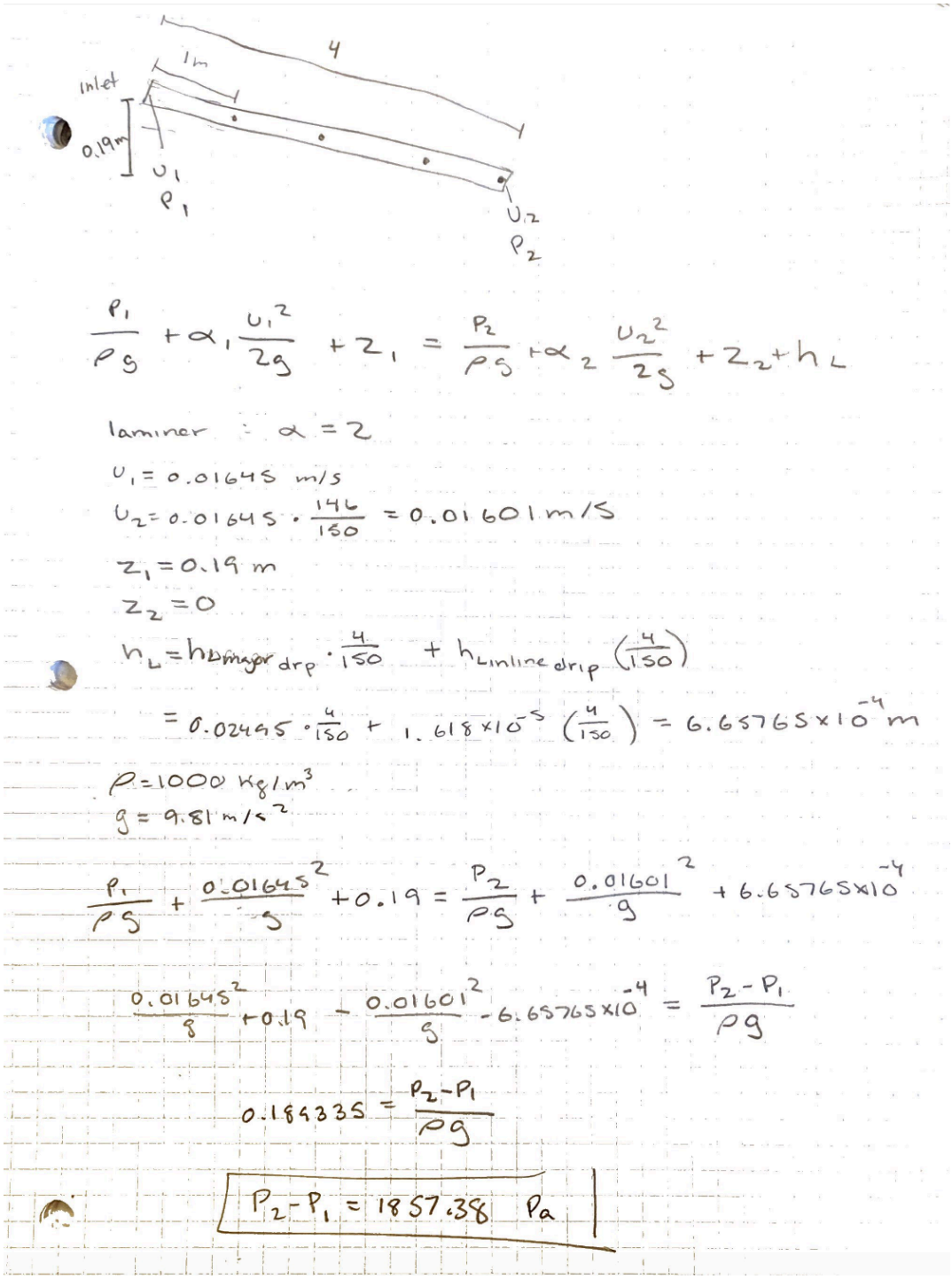
Table 1: *Percentage closure for each valve. Valve 1 is the valve closest to the PVC pipe, valve 9 is the valve closest to the outlet of the drip tape.*

Valve Number	P-Value (closure percentages)
1	94.98%
2	93.43%
3	91.64%
4	89.56%
5	87.05%
6	83.90%
7	79.68%
8	73.40%
9	61.42%

5) Numerical solution to compare with COMSOL®

Iteration 1

We chose to compare the pressure difference between the inlet and outlet of the modeled drip tape. For our analytical solution we assume that each of the 4 plants gets the ideal amount of water, meaning the outlet flow rate would be the inlet flow rate minus the water consumed by each plant. This assumption is reflected in the u_2 calculation.



Methods

Iteration 1

Geometry

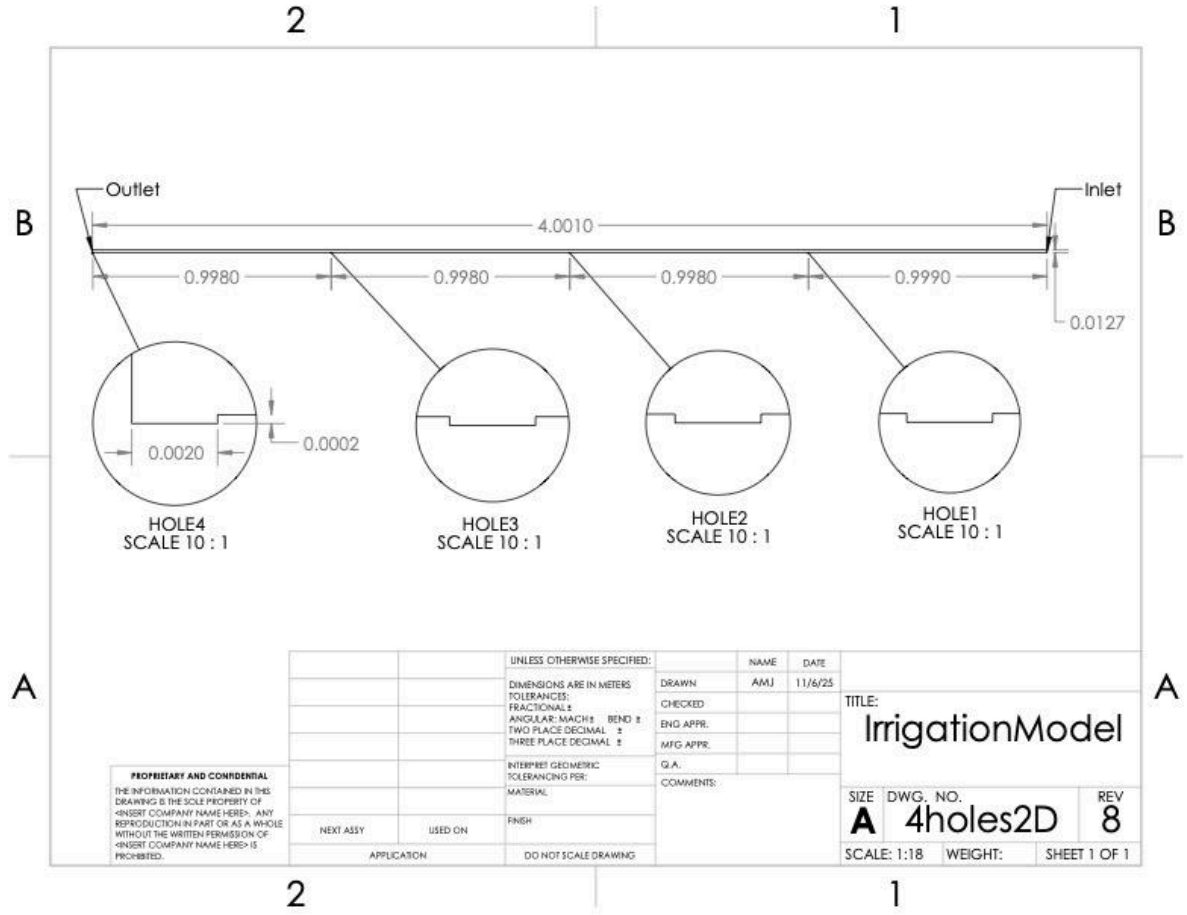


Figure 3: *Dimensioned Technical Drawing of Iteration 1 Model Geometry*

Our simulation models the beginning portion of the last column of eggplants in the field. It supplies water to four eggplants (i.e. we are modeling the first four outlets of the last column) from the larger-diameter PVC pipe (not included in the model). Figure 3 indicates where flow enters the geometry at the last elbow connecting the PVC to the drip tape. Note that the outlet label in the figure does not indicate where flow exits the drip tape since this is not the end of the column.

This geometry simulates its region of interest well with similar complexity and dimensions. However, this geometry only simulates a very small portion of the overall irrigation system. We simplified the geometry since modeling the field in its entirety would require having 22,500 holes supplying water to plants, which would take a lot of computational power to simulate in COMSOL®. This geometry allows us to analyze the flow each plant is receiving to determine the effectiveness of the irrigation system and allows us to see how multiple hole outlets interact with each other to affect the bulk flow speed and pressure.

Model

The Reynolds number for flow through a round pipe is calculated:

$$Re = \frac{Du_{avg}}{\nu} = (0.0127m \cdot 0.008225\frac{m}{s}) \div (1 \times 10^{-6}\frac{m^2}{s}) = 104.45$$

Thus, we elected to use the laminar flow model since we are analyzing an internal flow with Reynolds number well below 2,000. Gravity was included by using the slope of the field's incline (2.75125°) and calculating the component of gravity pointing in the direction of our sloped drip tape. In COMSOL®, this translated to adding a component of gravity in the x-direction with gravitational acceleration equal to $g \cdot sin(2.75125°)$.

Boundary Conditions

For our model, we used an inlet velocity of 0.01645m/s, calculated based on inlet flow rate which was assumed to be equal to the total flow rate required by the plants column. The four holes are outlets open to atmospheric pressure. Since we are not modeling an entire column of drip tape, the end of the segment of drip tape in our model is not open to the atmosphere but rather connected to more drip tape. For simplification, we modeled this as an open boundary, which simulates an interface with a large volume of fluid since this end of the tape is not exposed to the atmosphere like the holes.

Mesh

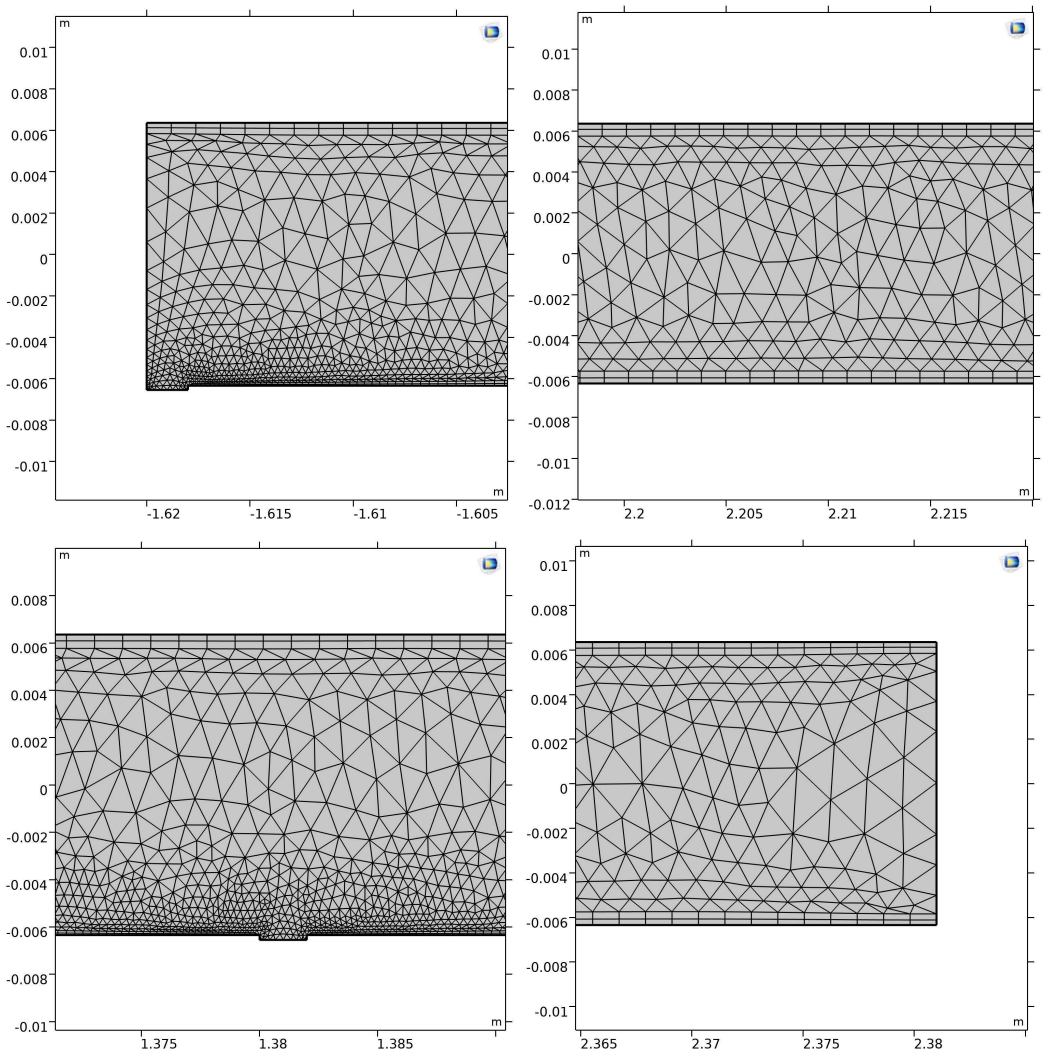


Figure 4: *Views of the iteration 1 model's mesh from left to right (outlet to inlet).*

To create our mesh, we used one of COMSOL®'s physics-controlled meshes with a 'normal' element size. This mesh consisted of 89,847 domain elements and 7,157 boundary elements (97004 elements total).

Results and Validation
Iteration 1

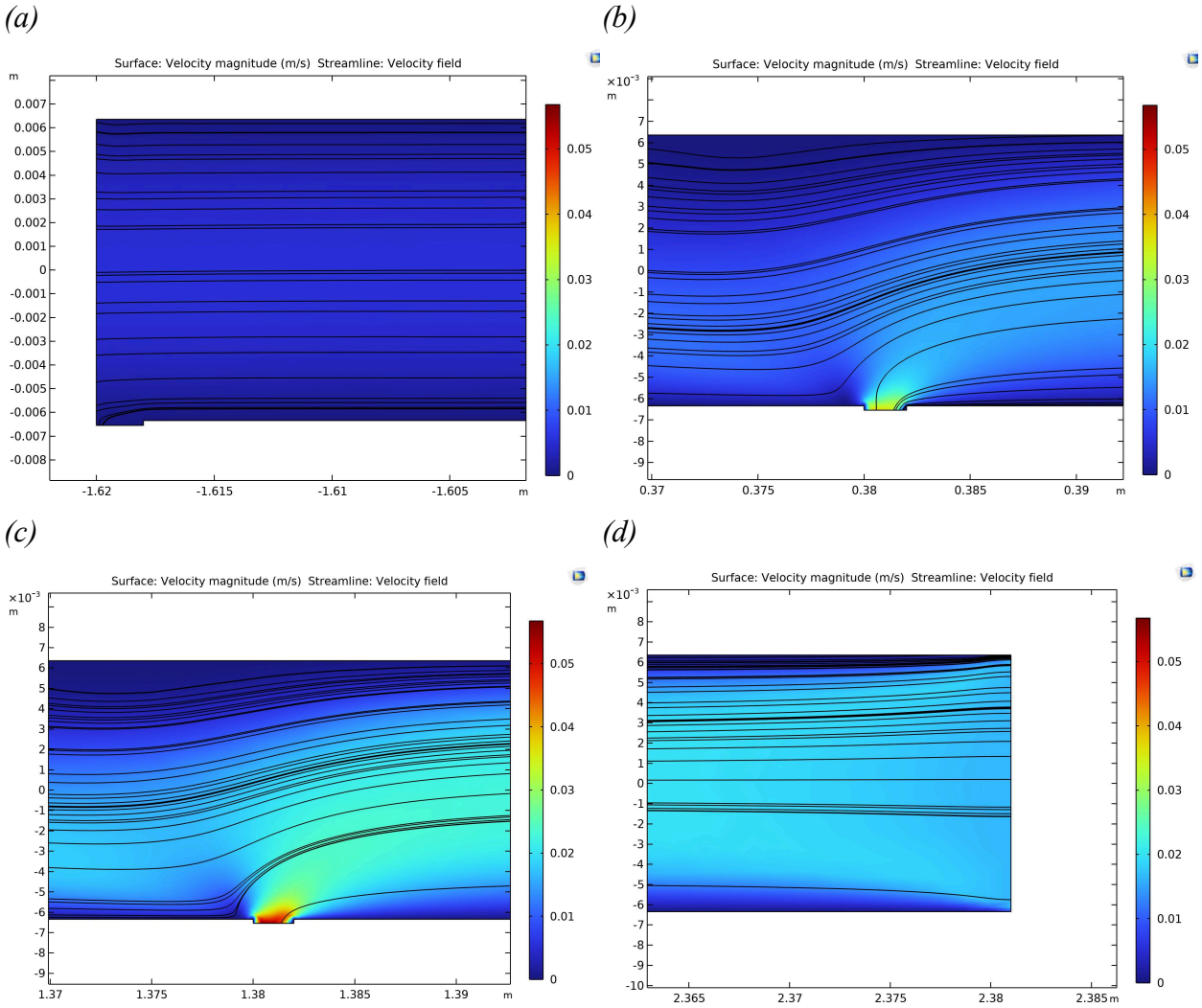


Figure 5: *Velocity streamline plots along the driptape. (a) hole 4 and outlet, (b) hole 2, (c) hole 1, (d) inlet.*

Plots (b) and (c) of *Figure 5* show that as flow exits the holes, it increases speed to fit through the small cross sections of the holes. A small separation region is visible on the opposite side of the tape slightly after each hole. The streamlines and color gradient on plot (d) show the increasing boundary layers and flow development at the inlet as streamlines move towards the center of the tape and the difference in velocity gradient develops moving away from the inlet. The color difference between plot (a) and (d) shows that the water is moving much faster at the inlet than the outlet.

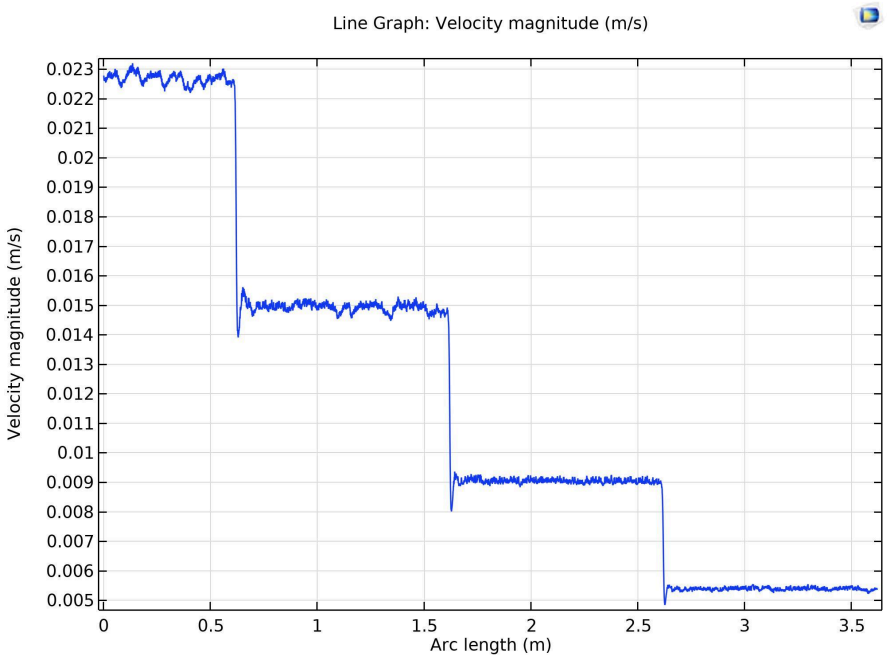


Figure 6: *Velocity magnitude (m/s) vs. distance (m) from the inlet along the centerline of the drip tape.*

The velocity plot shows that there is a large drop in velocity after every hole. These drops align with the drop in flow rate that comes from water exiting the holes. After each hole there is a slight recovery in velocity as flow smooths out after the hole disruption followed by pretty constant between holes.

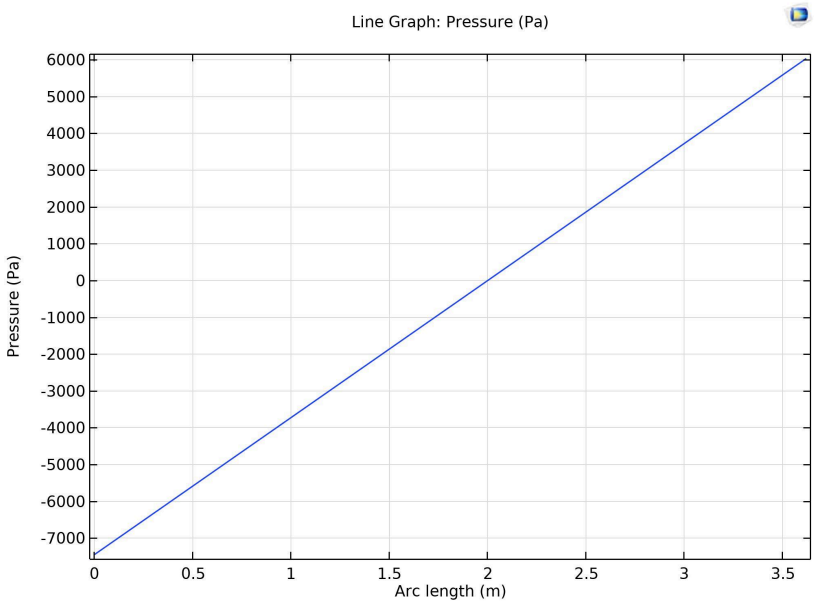


Figure 7: *Pressure (Pa) vs. distance (m) from the inlet along the centerline of the drip tape*

Figure 7 shows linear pressure increase along the center of the drip tape with a pressure of about -8000 Pa at the inlet and 6000 Pa at the outlet.

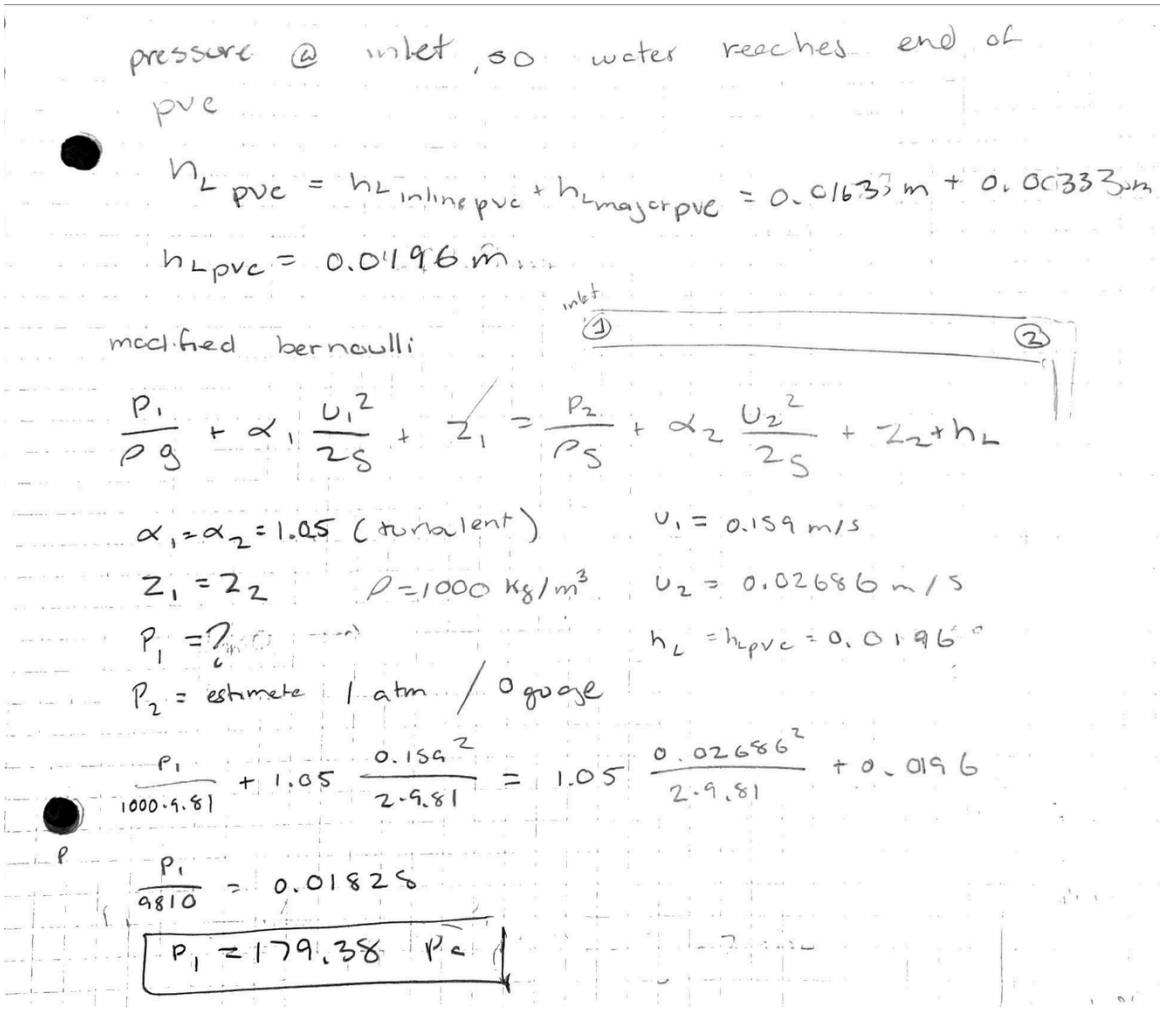
Quantitative analysis

We chose to compare the pressure change between the inlet and outlet of the modeled drip tape section. Our analytical solution showed this pressure change (P_{outlet}- P_{inlet}) to be 1857 Pa. The simulation indicates (P_{outlet}- P_{inlet}) is 14900 (P_{inlet}= -8866 Pa and P_{outlet}= 6034 Pa). These values have only a low level of agreement. The direction of pressure change is the same in both the analytical solution and the simulated solution with pressure increasing down the tube, so that outlet pressure is higher than inlet pressure. The simulation pressure change is approximately 8 times the analytical pressure change. One main discrepancy between the analytical solution and the simulated results is the assumption that each plant gets the ideal flow rate. This assumption is not realistic without the implementation of valves to control flow rate. An outlet velocity of 0.01601 m/s was used for the analytical solution while the simulated outlet velocity was approximately 0.0054 m/s. The analytical outlet velocity is approximately 3 times the simulated outlet velocity. Thus the discrepancy of outlet velocity squared is 9. Nevertheless the overall magnitude of the terms containing velocity are much smaller than the height terms, and have minimal impact on the final solution. One simplification we made for COMSOL® that may have impacted our results was the boundary condition put on the outlet. We modeled the outlet as an open boundary which assumes it interfaces with a large volume of fluid. While the actual system does interact with fluid at this point, it interacts with the continued flow of the drip pipe, not a large volume. This simplification may have impacted COMSOL®'s pressure assumptions for this point and the overall pressure change.

Sociotechnical Questions

Water Pressure and Pump

Based on our calculations, an inlet pressure of 179.38 Pa is necessary for the flow to reach the end of the PVC pipe, reaching every drip tape for the field. This is 0.026 psi, which the units that water pumps are more commonly sold in. This is quite a low value for a water pump, and the farmers will not be able to find any pumps which supply only 0.05 psi or so.



This means that a farmer has their pick of any pump for their system. Our considerations when looking for pumps to pick were price, reliability, and capability. The Institute for Poverty Action states that most farmers in Zambia make less than $500 per year, and many struggle with only getting paid once or twice a year, due to the nature of crop harvests and sale.[5] Home Depot sells a power head pond and fountain pump for $14.[6] It is made by the Alpine Corporation, and is generally reviewed well by customers. Since this pump would not be used at its maximum

[5]https://poverty-action.org/study/impact-advanced-payment-farmer-wellbeing-and-company-profitability-zambiahttps://poverty-action.org/study/impact-advanced-payment-farmer-wellbeing-and-company-profitability-zambia

[6]https://www.homedepot.com/p/Alpine-Corporation-Outdoor-Decor-Power-Head-Pump-65-GPH-for-Large-Fountains-Ponds-and-Birdbaths-P065/206515145

capacity, we feel more confident that it would last longer than it otherwise might. The pump is lightweight and simple to use. Even if the pump does dip below its advertised 65 gallons per hour, it will still be well above the needs of the farm, since at its max, it pumps at around 120,040 Pa, while this system only requires 180 Pa.

Blockage

Preventing blockages from causing widespread harm was one of the biggest considerations of our initial design. Folotiya speaks about the impact of foulings on his YouTube channel, citing sediment build up or dirt entering the system as two of the largest risk factors. Our design is made such that there is a lot of redundancy without adding unnecessary complexity which would result in increased head loss and extra setup effort or expense for a farmer. The PVC feeder pipe is significantly larger in diameter than the drip tape, and is therefore unlikely to be blocked. Each drip tape line emerges directly from the PVC, heading down the field. The drip tape is the weakest point for a blockage, since it is in the dirt, and also because it has a significantly smaller pipe diameter and outlet diameters than any other part of the irrigation system. This design also makes clear where in the system a fouling takes place. A farmer can work backwards along the grid system of plants to determine where the water flow to the eggplants ends. This significantly cuts down on the searching time for blockages and allows the farmer to fix them in a timely manner, reducing the harm to crops and profit lost. We used COMSOL® to model a blockage in the middle of a pipe, the results of which are seen in *Figure 10*.

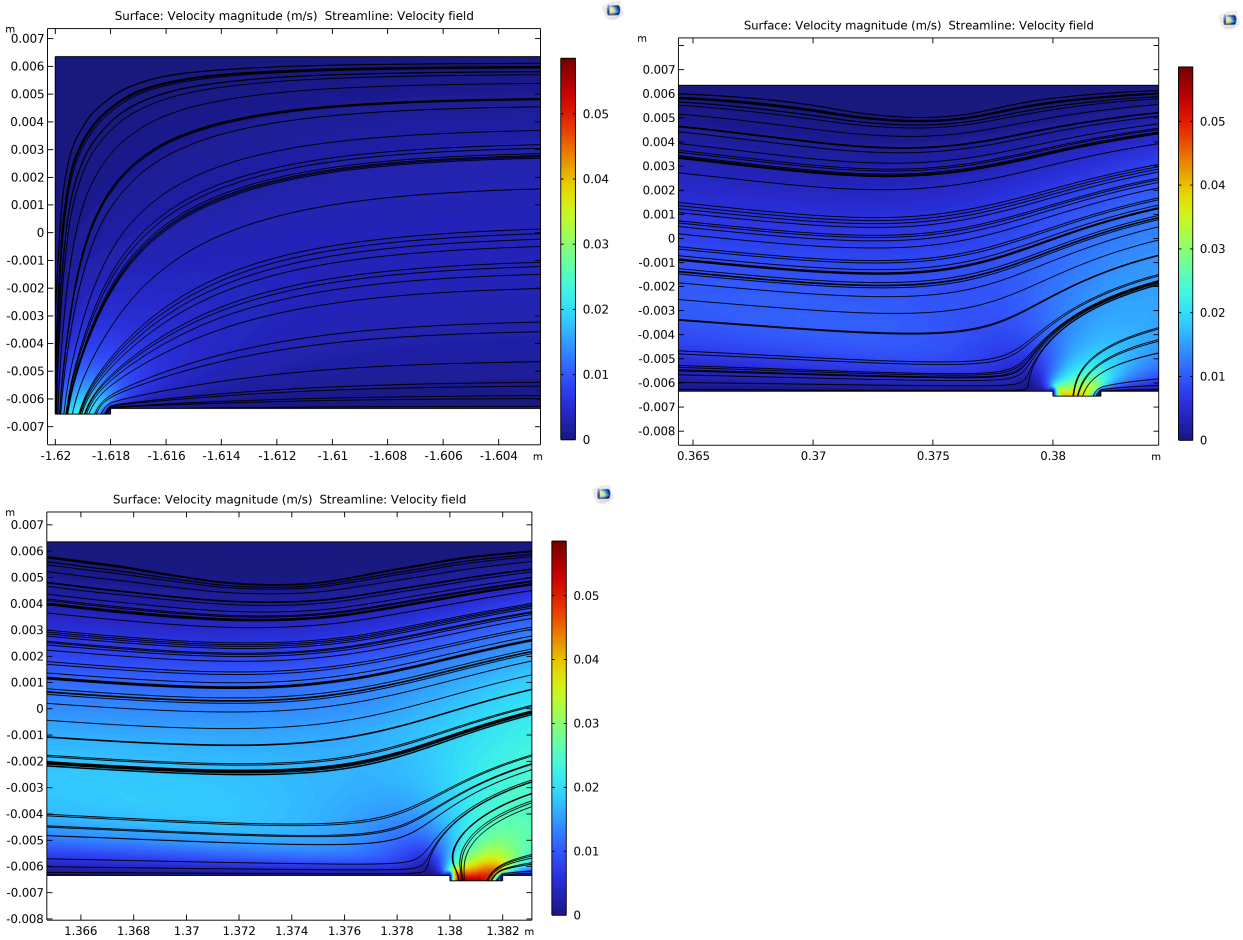


Figure 8: *Velocity streamline plots along the drip tape with a blockage.*

Adding a blockage after the four holes in our model indicates that these four holes before the blockage will have higher outlet velocities (particularly for the downstream holes) since flow is completely cut off to the downstream portion of the drip tape. For example, with the introduction of the blockage the outlet velocity at hole 1 increases from 0.0568 to 0.0586 m/s, at hole 2, it increases from 0.0376 to 0.0411 m/s and at the end, it increases from 0.000368 to 0.0218 m/s, showing that the change in flow is more significant closer to the blockage.

The worst realistic fouling would be a blockage at the very start of the drip tape, right where it meets the PVC feed pipe. This would cut off one full column of eggplants from receiving water. If not fixed in time, this would result in the death of 150 plants. If each plant produces 2-4 eggplants[7], this would result in a worst case scenario of 600 eggplants lost, 10% of total expected crop yield. If each eggplant weighs 1 kg[8], that's 600 kgs of lost eggplant, and at current rates of $2.00 USD per eggplant, that's a loss of a maximum of $1,200 for the farmer. This loss is before even factoring in the expense of replacing and replanting the lost plants.[9]

[7] https://plantvillage.psu.edu/posts/2886-eggplant-how-many-eggplants-should-i-plant
[8] https://en.wikipedia.org/wiki/Eggplant
[9] https://www.selinawamucii.com/insights/prices/zambia/aubergines/

Redundancy

Redundancy is a hard thing to redesign for since our initial design was so focused on safety in case of a blockage. Other than potentially splitting every drip tape flow into two drip tapes covering the exact same region, so that if one clogs the other will still supply water, we don't see many efficient possibilities for improving our design. If we were to split into two drip tapes, that effectively doubles our chance of a blockage since there is twice the number of holes and twice the number of drip tapes to become blocked, so we end up right where we started after a significant amount of extra setup and cost.

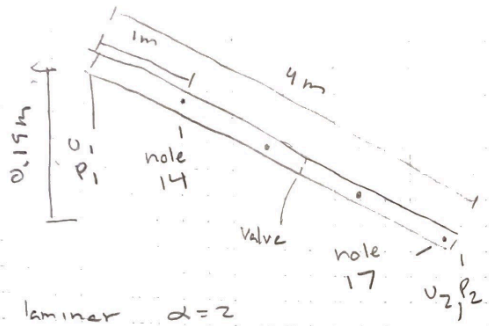
Iteration 2

Updated Design

The main goal of the second simulation was to create a more representative model of the irrigation system by including the valves in the system. We simulated valve 1, which closes off 94.98% of the drip tape opening, as seen in Table 1, which works to normalize the flow rate throughout the rest of the tape.

Analytical solution

We added a valve to our COMSOL® iteration for the second iteration comparison of pressure change. To analyze the valve, the 4 plants surrounding the first valve were analyzed (plants 14, 15, 16, and 17 from the top of the drip tape). This change is reflected in the u_1 and u_2 calculations. For our analytical solution, we assume that each of the 4 plants gets the ideal amount of water, meaning the outlet flow rate would be the inlet flow rate minus the water consumed by each plant. This assumption is reflected in the u_2 calculation. There was negligible change to the final analytical solution for pressure difference.



laminar $\alpha = 2$

$$v_1 = 0.01645 \cdot \frac{137}{150} = 0.015024 \text{ m/s}$$

$$v_2 = 0.01645 \cdot \frac{133}{150} = 0.014586 \text{ m/s}$$

$z_{1\,rel} = 0.19\text{m}$

$z_{2\,rel} = 0\text{ m}$

$$h_{L\,valve} = 137 e^{-8.3\,(0.95)} \frac{0.015024^2}{2g}$$

$$h_{L\,valve} = 5.93198 \times 10^{-7}$$

$$h_L = h_{L\,major\,drop} \frac{4}{150} + h_{L\,minor\,drop} \left(\frac{4}{150}\right) + h_{L\,valve}$$

$$h_L = 0.02495 \cdot \frac{4}{150} + 1.618 \times 10^{-5} \left(\frac{4}{150}\right) + 5.931984 \times 10^{-7}$$

$h_L = 6.6636 \times 10^{-4}\text{m}$

$\rho = 1000\ kg/m^3$

$g = 9.81\ m/s^2$

$$\frac{P_1}{\rho g} + \alpha_1 \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{v_2^2}{2g} + z_2 + h_L$$

$$\frac{P_1}{\rho g} + \frac{0.015024^2}{g} + 0.19 = \frac{P_2}{\rho g} + \frac{0.014586^2}{g} + 6.6636 \times 10^{-4}\text{m}$$

$$\frac{P_2 - P_1}{\rho g} = 0.1843349$$

$P_2 - P_1 = 1857.38\ Pa$

Geometry

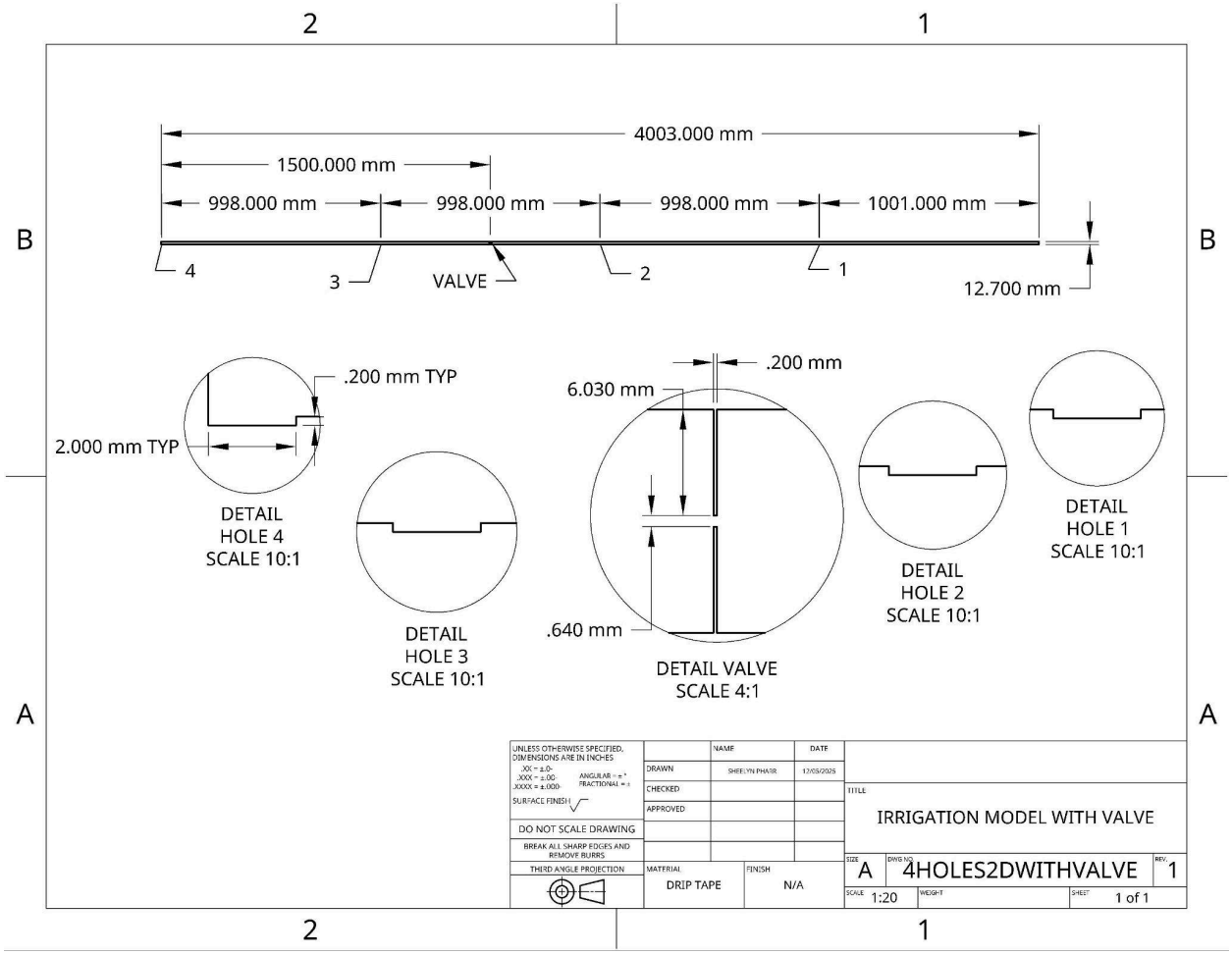


Figure 9: *Dimensioned Technical Drawing of Iteration 2 Model Geometry*

Our simulation models the first valve and surrounding plants of the last column of eggplants in the field. It supplies water to eggplants 14, 15, 16, 17 (i.e. we are modeling two plants before the valve and two plants after the valve). The larger-diameter PVC pipe is not included in the geometry. Figure 5 indicates where flow enters the geometry after plant 13, 13 meters down the driptape from the transition form PVC. Note that the outlet label in the figure does not indicate where flow exits the drip tape since this is not the end of the column.

This iteration of geometry simulates the valve not included in the first iteration. The valve is simulated with 2 walls covering a total of 95% of the cross section and leaving an opening in the middle. It simulates its region of interest well with similar complexity and dimensions. However, this geometry only simulates a very small portion of the overall irrigation system.

Model

The Reynolds number for flow through a round pipe is calculated:

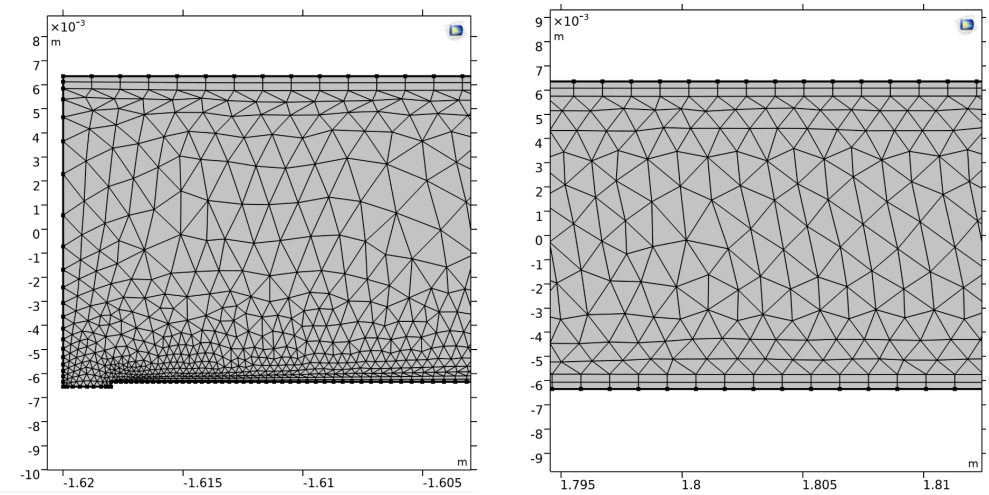
$$Re = \frac{Du_{avg}}{\nu} = (0.0127m \cdot 0.008225\frac{m}{s}) \div (1 \times 10^{-6}\frac{m^2}{s}) = 104.45$$

Thus, we elected to use the laminar flow model since we are analyzing an internal flow and with Reynolds number well below 2000. Gravity was included by using the slope of the field's incline (2.75125°) and calculating the component of gravity pointing in the direction of our sloped drip tape. In COMSOL®, this translated to adding a component of gravity in the x-direction with gravitational acceleration equal to $g \cdot sin(2.75125°)$. These conditions are the same as the first iteration.

Boundary Conditions

For our model, we used an inlet velocity of 0.015024 m/s, calculated based on inlet flow rate to the drip pipe and the ideal amount of water taken by the 13 plants above the region of focus on the drip tape. The four holes are outlets open to atmospheric pressure. Since we are not modeling an entire column of drip tape, the end of the segment of drip tape in our model is not open to the atmosphere but rather connected to more drip tape. For simplification, we modeled this as an open boundary, which simulates an interface with a large volume of fluid since this end of the tape is not exposed to the atmosphere like the holes.

Mesh



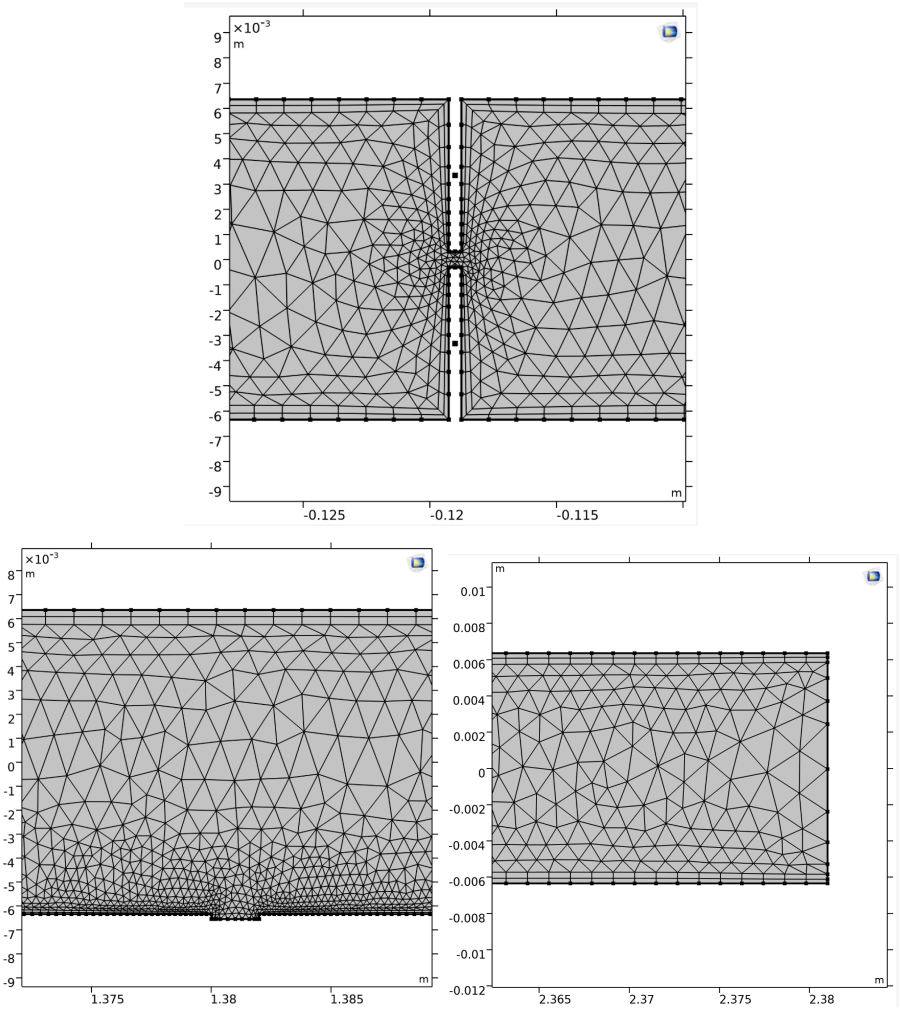
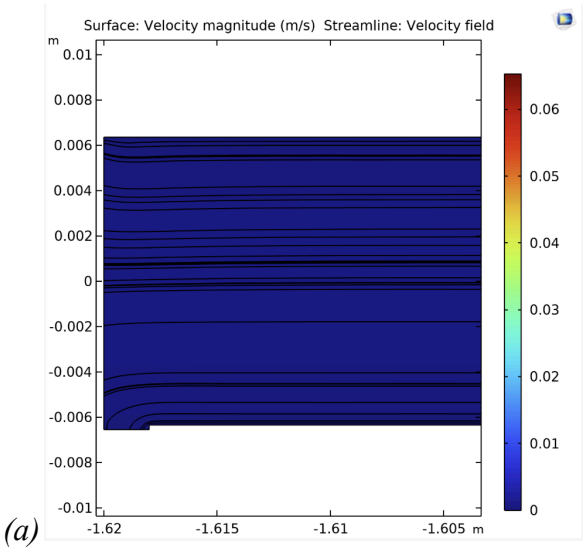


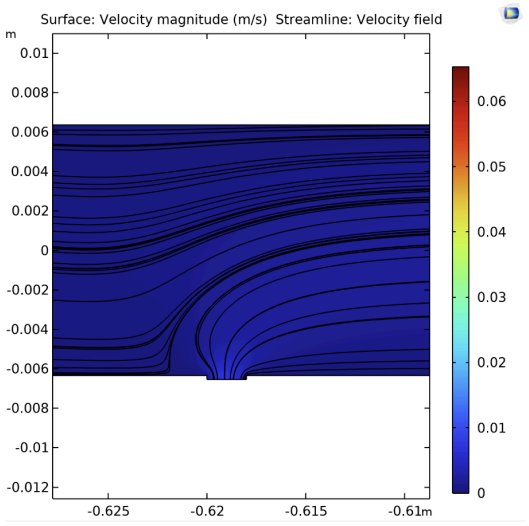
Figure 10: *Views of the iteration 2 model's mesh from left to right (outlet to inlet).*

To create our mesh, we used one of COMSOL®'s physics-controlled meshes with a 'normal' element size. This mesh consisted of 89,451 domain elements and 7,161 boundary elements (96612 elements total).

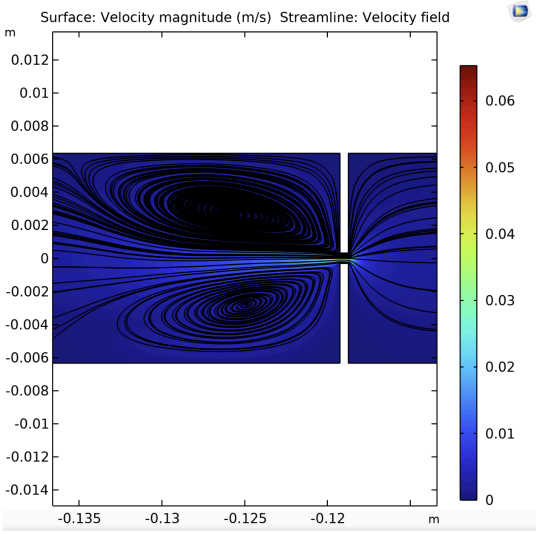
Iteration 2



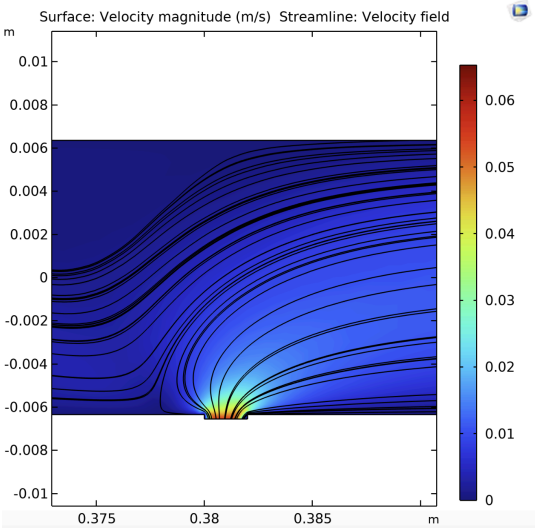
(a)



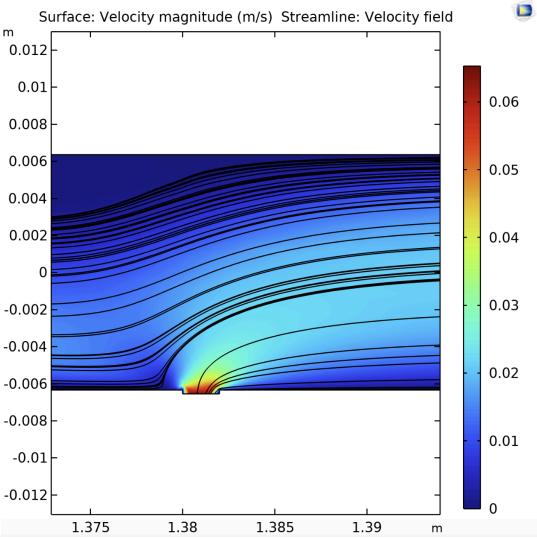
(b)



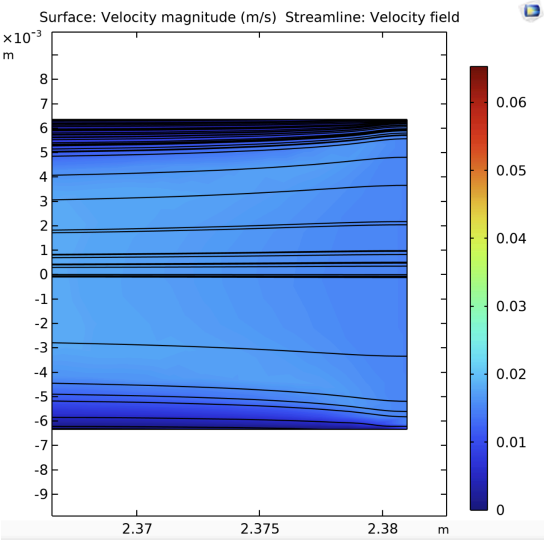
(c)



(d)



(e)



(f)

Figure 11: *Velocity streamline plots along the driptape. (a) hole 4 and outlet, (b) hole 3, (c) valve, (d) hole 2, (e) hole 1(f) inlet.*

Plots (b), (d), and (e) show that flow speeds up at the drip tape holes. Plots (d) and (e) show that before the valve the separation regions downstream of the holes are large and growing, but plot (b) shows that there is no separation region for the hole after the valve. Plot (c) of the valve shows an increase in velocity for flow to fit through the valve and lots of recirculation after the valve. The difference in color between plot (f) and (a) indicates that flow slows down along the drip tape.

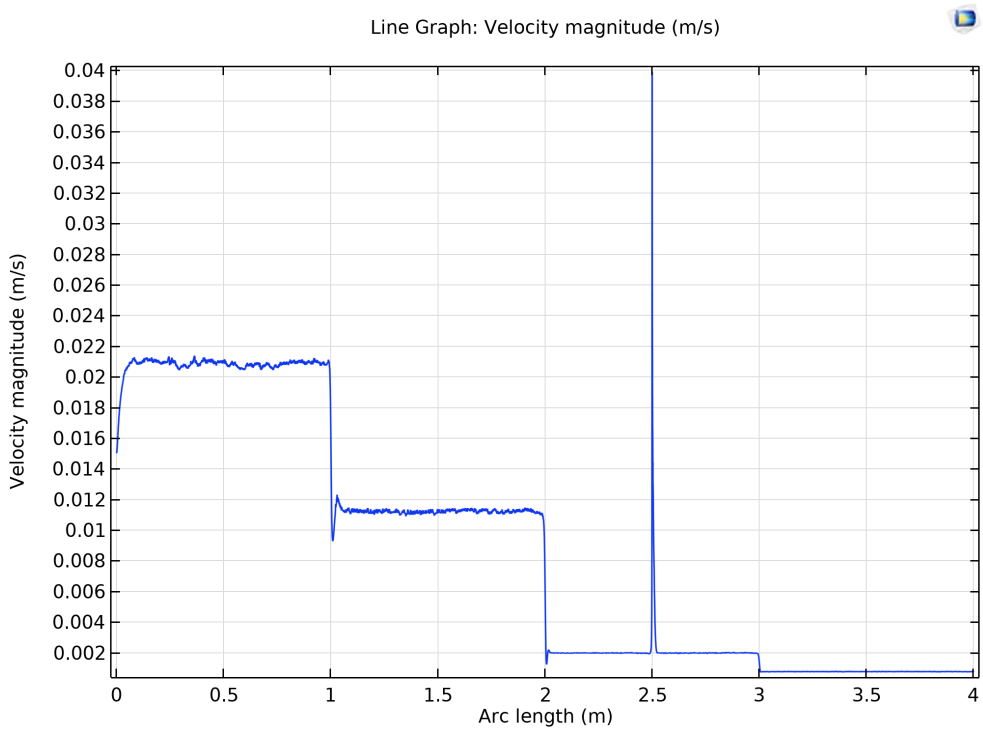


Figure 12: *Velocity magnitude (m/s) vs. distance (m) from the inlet along the centerline of the drip tape.*

The initial increase in velocity visible in Figure 12 indicates that at the inlet the flow quickly accelerates before reaching a constant speed before the first hole. Each hole results in a drop in velocity with a slight dip after the hole as flow recovers. The valve causes a major spike in velocity at arc length 2.5 m due to the sudden decreased cross-sectional area for flow. Furthermore the change in velocity after hole 3 is much less than after holes 1 and 2. Due to the significant energy loss concentrated at the valve, each subsequent outlet removes only a small fraction of an already-small flow, so the velocity decreases only slightly after each hole compared to the holes before the valve.

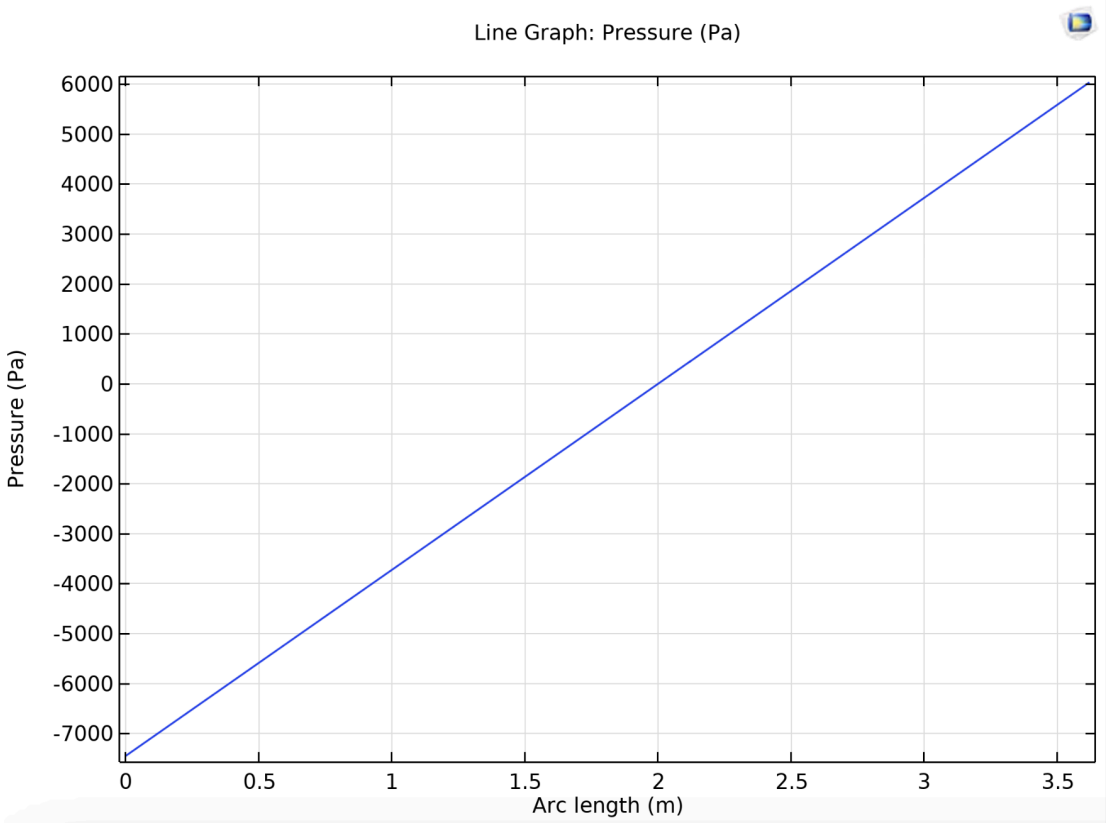


Figure 13: *Pressure (Pa) vs. distance (m) from the inlet along the centerline of the drip tape*

Figure 13 shows linear pressure increase along the center of the drip tape with a pressure of about -8000 Pa at the inlet and 6000 Pa at the outlet. This trend closely resembles Figure 7 of the iteration 1 model.

Quantitative analysis

Our analytical and COMSOL® solution of pressure change has the same low level of agreement as they did in iteration 1. Our analytical solution showed this pressure change (P_{outlet}- P_{inlet}) to be 1857 Pa. The simulation indicates (P_{outlet}- P_{inlet}) is 14900 (P_{inlet}= -8865 Pa and P_{outlet}= 6034 Pa). The pressure difference calculated from the COMSOL® solution is 8 times the difference from our analytical solution. The boundary condition assumptions used may have contributed to this variation. We simulated the outlet as an open boundary condition in COMSOL® when in reality the flow continues through the remainder of the drip tape. This variation may impact the flow at the simulated outlet, increasing the pressure difference. Outlet velocities as seen in the analytical solution, Figure 11, and COMSOL® probing indicate discrepancy between the COMSOL® outlet velocity and the analytical outlet velocity, with magnitudes to the power of negative 4 and negative 2 respectively. This discrepancy supports the assumption that outlet conditions of the COMSOL® model are a large factor in the disagreement of our quantitative parameter.

Design Reflection

The main goal of the second simulation was to create a more representative model of the irrigation system by including the valves in the system. We simulated valve 1, which closes off 94.98% of the drip tape opening, which works to normalize the flow rate throughout the rest of the tape. Our second iteration did not improve the quantitative comparison with our analytical solution, but gave us a more accurate view of the flow's movement. There were negligible changes in the pressure difference between iteration 1 and 2 for the analytical or simulated solutions. The lack of improvement can be attributed to the change in height of driptape on the slanted ground overpowering the additional headloss from the valve. The change in z was about one million times bigger than the headloss from the valve. Despite the lack of change in the quantitative comparison, we saw major changes in the velocity plots. As seen in *Figure 12*, the valve created a large spike in velocity as the flow tried to squeeze through the small opening. As seen in *Figure 11* there is a region of recirculation after the valve. In plots (d) and (e) of *Figure 11* the separation region following hole 2 is larger than the separation region following hole 1. This trend is also visible in plot (b) and (c) of *Figure 8*, but is exaggerated in *Figure 11* because both separation regions are larger in the second iteration. If we were to do more iterations of the simulation, we would like to try to get COMSOL® to simulate bigger setups. We were only able to model a fairly small portion of the drip tape, which did give us fairly representative values, but it would be interesting to model more of the tape or to model the connection from the PVC pipe to the drip tape.

Social Reflection

Having Folotiya's youtube channel proved to be a crucial resource during our research phase of the project. Being able to see the farm in video really helped us inform our design and understand the factors which are most important to local farmers. It would be helpful to ask him about the positioning of the drip tape holes and whether they face down, to the side, or up. We also relied on many sources about disaster outreach, such as the United Nations, to learn about the impacts of the drought. It would have been nice to speak with more farmers, in Zambia or in the United States, to learn about their concerns with adopting a drip tape irrigation system or to learn about more problems drip tape users run into that we are unaware of. We would want to ask about what part of their current irrigation systems is their least favorite and what is their favorite, we would also ask about the cost of the water they use to irrigate their crops, and if they think reducing their water usage is worth it. We would also like to ask farmers and non-farmers in Zambia about the potentially unseen impacts of the drought that they experience.

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